

1. COMPUTING WITH REAL NUMBERS

Floating-point system

Normalized binary FPN:  $\text{fl}(x) = \pm(1.d_1d_2\dots d_p)_2\,2^e$ ,  $d_i \in \{0,1\}$ ,  $e_{\min} \leq e \leq e_{\max}$ . IEEE double: sign 1 bit, p=52 fraction bits, 11 exponent bits; normal  $e \in [-1022, 1023]$ .  $\epsilon = 2^{-52} \approx 2.22 \times 10^{-16}$  (gap from 1 to next FPN);  $e_i = 2^m e$  ( $2^m \leq i < 2^{m+1}$ ),  $x_{\max} = (2 - \epsilon)2^{1023}$ ,  $x_{\min}^{\text{normal}} = 2^{-1022}$ . Below: subnormal, then underflow to 0; overflow  $\rightarrow \pm\infty$ ; undefined  $\rightarrow \text{NaN}$ . ***FPN arithmetic is not associative; avoid exact equality tests.***

**Errors, accuracy, stability**

For approximation  $\tilde{x}$  to  $x$ :  $e_a = |\tilde{x} - x|$ ,  $e_r = \frac{|\tilde{x} - x|}{|x|}$  ( ***$x \neq 0$*** ).

Significant decimal figures: largest p satisfying  $e_r < \frac{1}{2}10^{-p}$ . Storage rounding error  $\leq \epsilon|x|$ . **Catastrophic cancellation:** subtracting computed nonzero  $\tilde{x}, \tilde{y}$  when  $x \approx y$  can greatly amplify relative error. Quadratic  $ax^2 + bx + c = 0$ :  $q = -\frac{1}{2}(b + \text{sign}(b)\sqrt{b^2 - 4ac})$ ,  $x_1 = \frac{q}{a}$ ,  $x_2 = \frac{c}{q}$ . ***Require  $a \neq 0$ , real-root form  $b^2 - 4ac \geq 0$ ,  $q \neq 0$  for  $c/q$ .***

Cost and storage

1 bit  $\in \{0,1\}$ ; 1 byte = 8 bits; KiB =  $2^{10}$  bytes, MiB =  $2^{20}$ , GiB =  $2^{30}$ . Signed b-bit integer range  $[-2^{b-1}, 2^{b-1} - 1]$ . 1 flop = one FPN +, −, ×, /. Speed = (cycles/s)/(cores)(flops/cycle); time = flops/speed. memory( $n \times n$ ) =  $8n^2$  bytes, flops( $AB$ ) =  $n^2(2n - 1) \approx 2n^3$ .

2. NORMS, EQUALITY, SENSITIVITY

Vector norms and error

For  $x \in \mathbb{R}^n$ ,  $p \geq 1$ :  $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}$ ;

$\|x\|_1 = \sum_i |x_i|$ ,  $\|x\|_2 = \sqrt{x^T x}$ ,  $\|x\|_\infty = \max_i |x_i|$ . Norm axioms:

positivity/definiteness, triangle inequality, absolute homogeneity.  $x = y \iff \|x - y\| = 0$ ,  $\|x - y\| \leq \tau(e, n, \|x\|, \|y\|)$ .

$e_a = \|\tilde{x} - x\|$ ,  $e_r = \frac{\|\tilde{x} - x\|}{\|x\|}$  ( ***$x \neq 0$*** );

$\frac{\|\tilde{x} - x\|_\infty}{\|x\|_\infty} < \frac{1}{2}10^{-k}$ ,  $k = \lfloor -\log_{10}(2e_r, \infty) \rfloor$ .

Matrix norms

Subordinate norm:  $\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}$ ;

$\|Ax\|_p \leq \|A\|_p \|x\|_p$ ,  $\|AB\|_p \leq \|A\|_p \|B\|_p$ .

$\|A\|_1 = \max_j \sum_i a_{ij}$ ,  $\|A\|_\infty = \max_i \sum_j a_{ij}$ ,  $\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sigma_1$ ;

Frobenius  $\|A\|_F = \sqrt{\sum_{ij} |a_{ij}|^2}$  (***not generally  $\|A\|_2$*** ).

Conditioning of  $Ax = b$

For nonsingular A:  $\kappa(A) = \|A\| \|A^{-1}\| \geq 1$ ; rcond =  $1/\kappa$ .

$\kappa_2(A) = \frac{\max_i |x_i|}{\min_i |x_i|}$  ( $A = A^T$ ),  $\kappa_2(A) = \frac{\sigma_1}{\sigma_n}$  ( $\text{rank}(A) = n$ ).

$(A + \delta A)(x + \delta x) = b + \delta b$ ,  $\|(A + \delta A)^{-1}\| \leq \alpha \|A^{-1}\|$ ;  $A + \delta A$

nonsingular,  $\alpha > 1$ .  $\rho_A = \frac{\|\delta A\|}{\|A\|}$ ,  $\rho_b = \frac{\|\delta b\|}{\|b\|}$ ,  $\frac{\|\delta x\|}{\|x\|} \leq \alpha \kappa(A)(\rho_A + \rho_b)$ .

Large  $\kappa \Rightarrow$  sensitive/ill-conditioned;  $\kappa \geq 1/\epsilon$  can destroy machine-precision information.

3. MATRIX STRUCTURE & FACTORISATIONS

Special matrices

$C_{ij} = E[(X_i - \mu_i)(X_j - \mu_j)]$ ,  $C_{ii} = \sigma_i^2$ ; C symmetric PSD.

$K_{ij} = \frac{C_{ij}}{\sigma_i \sigma_j}$ ,  $K = \Sigma^{-1} C \Sigma^{-1}$ ,  $K_{ii} = 1$ ,  $|K_{ij}| \leq 1$  ( ***$\sigma_i, \sigma_j > 0$*** ).

Upper/lower triangular:  $a_{ij} = 0$  for  $i > j$  /  $i < j$ .  $\det A = \prod_i a_{ii}$ , A nonsingular  $\iff a_{ii} \neq 0 \, \forall i$ . Unit triangular:  $a_{ii} = 1$ .

Triangular solve  $\approx n^2$  flops.

$a_{ij} = 0$  if  $i - j > m_\ell$  or  $j - i > m_u$ ; total bandwidth  $m_\ell + m_u + 1$ ;  $\text{nnz} \leq (1 + m_\ell + m_u)n - m_\ell(m_\ell + 1)/2 - m_u(m_u + 1)/2$ . Tridiagonal:  $m_\ell = m_u = 1$ . Toeplitz:  $a_{ij} = \alpha_{j-i}$ .

Symmetric  $A^T = A$ ; skew  $A^T = -A$ .  $A = \frac{A + A^T}{2} + \frac{A - A^T}{2}$ . Definiteness

by  $x^T A x$ : PD  $> 0$ , PSD  $\geq 0$ , ND  $< 0$ , NSD  $\leq 0$ ; for real symmetric A, signs of eigenvalues give type. Strict row diagonal dominance:  $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ ; symmetric + strict dominance  $\Rightarrow$  PD.

$Q^T Q = I$ ,  $Q^{-1} = Q^T$ ,  $\kappa_2(Q) = 1$ ,  $\|Qx\|_2 = \|x\|_2$ .

Permutation matrices are orthogonal.

LU and Cholesky

LU:  $A = LU$  with L unit lower, U upper; when it exists for nonsingular A it is unique. In general partial pivoting gives  $PA = LU$ ;  $|\ell_{ij}| \leq 1$ . Factor cost  $\approx 2n^3/3$ ; two triangular solves  $\approx 2n^2$ . ***Require nonsingular A for a unique solve; do not form  $A^{-1}$ .***

Cholesky:  $A = R^T R$ , R upper. ***Require real symmetric positive-definite A***; factor cost  $n^3/3$ , solves  $2n^2$ ; no row pivoting.

QR

For  $A \in \mathbb{R}^{m \times n}$ ,  $m \geq n$ ,  $\text{rank}(A) = n$ :

$A = Q \begin{bmatrix} R \\ 0 \end{bmatrix} = YR$ ,  $Q = [Y \, Z]$ ,  $Q^T Q = I$ ; R nonsingular upper.

Cost  $\approx 2mn^2 - (2/3)n^3$ . If rank deficient:  $AP = Q \begin{bmatrix} R \\ 0 \end{bmatrix}$ , column pivoting,  $|r_{11}| \geq \dots \geq |r_{nn}|$ ; small diagonal entries indicate numerical rank deficiency.

SVD

$A \in \mathbb{R}^{m \times n}$ ,  $m \geq n$ :  $A = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^T$ ; U, V orthogonal,

$\Sigma = \text{diag}(\sigma_1 \geq \dots \geq \sigma_n \geq 0)$ .  $\text{rank}(A) = \text{number of positive } \sigma_i$ ;

$A = \sum \sigma_i u_i v_i^T$ ,  $A v_i = \sigma_i u_i$ ,  $u_i^T A = \sigma_i v_i^T$ . Full rank:  $\kappa_2(A) = \sigma_1/\sigma_n$ .

Numerical rank tolerance  $\approx \max(m, n)\epsilon \sigma_1$ .

4. PARAMETRIC DATA FITTING

Residual model

Data  $(t_i, y_i)$ ,  $i = 1, \dots, m$ ; parameters  $x \in \mathbb{R}^n$ ,  ***$m \geq n$*** .  $r_i(x) = \phi(x; t_i) - y_i$ ,  $x^* = \arg \min_x \|r(x)\|_p$ . Interpolation  $\Leftrightarrow r = 0$  (***appropriate only for highly accurate data***).

Linear-in-parameters:  $\phi(x; t) = \sum_j x_j \phi_j(t)$ ,  $A_{ij} = \phi_j(t_i)$ ,  $r = Ax - y$ .

Linear least squares (p=2)

Normal equations:  $A^T A x^* = A^T y$ ; residual orthogonal to  $\text{col}(A)$ .  $A^T A$  PSD, and PD iff A has full column rank. ***Avoid forming  $A^T A$  when conditioning matters:***  ***$\kappa_2(A^T A) = \kappa_2(A)^2$ .***

Pivoted QR  $AP = Q[R; 0]$ ,  $Q = [Y \, Z]$ :  $R(P^T x^*) = Y^T y$ . SVD

$A = U[\Sigma; 0]V^T$ ,  $U = [Y \, Z]$ :  $x^* = V\Sigma^{-1}Y^T y$ ,  $x_{\min}^* = V\Sigma^+ Y^T y$ . Use  $\Sigma^+ = \text{diag}(1/\sigma_1, \dots, 1/\sigma_r, 0, \dots)$  for  $\text{rank } r < n$ .

Choice of norm / transformation

p=2: suited to independent  $N(0, \sigma^2)$  errors with common variance. p=1: robust to outliers,  $\min \sum v_i$ :  $-v_i \leq r_i \leq v_i$ ,  $v_i \geq 0$ . p= $\infty$ :  $\min v$ :  $-v \leq r_i \leq v$ ,  $v \geq 0$ .

Linear r gives linear programs.

Log-linearisation: products/exponentials/powers may be transformed. ***Log transform requires every logged quantity positive and any shifted base  $t - \beta_j > 0$ .***

5. CUBIC SPLINES & SPARSE MATRICES

Polynomials and interpolation

$p(t) = \sum_{k=0}^n a_k t^k = a_0 + \ell_1 a_1 + \ell_2 a_2 + \dots + \ell_n a_n$  (Horner form)

uses n multiplies + n adds. Weierstrass: for continuous f on closed  $[a, b]$ ,  $\forall \epsilon > 0$  polynomial p with  $|f - p| < \epsilon$  on  $[a, b]$ . Polynomial interpolation at distinct  $t_0, \dots, t_n$ : Vandermonde  $Ax = y$ ,  $A_{ij} = t_i^j$ . ***High-degree equally spaced interpolation can be ill-conditioned and oscillatory; use low-degree pieces.***

Cubic spline

Knots  $a = t_0 < \dots < t_n = b$ .  $s(t) = s_i(t)$  on  $[t_{i-1}, t_i]$ , degree( $s_i$ )  $\leq 3$ , and  $s \in C^2[a, b]$ . Interpolation  $s_i(t_i) = y_i$ ; at interior knots  $s_i(t_i) = s_{i+1}(t_i)$ ,  $s_i'(t_i) = s_{i+1}'(t_i)$ ,  $s_i''(t_i) = s_{i+1}''(t_i)$ . Two endpoint conditions required:

natural:  $s''(a) = s''(b) = 0$ ; clamped:  $s'(a), s'(b)$ ; not-a-knot:  $s_i'''(t_1) = s_2'''(t_1)$ ,  $s_{n-1}'''(t_{n-1}) = s_n'''(t_{n-1})$ .

Uniform  $h = t_i - t_{i-1}$ ,  $d_i = (y_i - y_{i-1})/h$ ,  $M_i = s''(t_i)$ . For  $t \in [t_{i-1}, t_i]$ :  $s_i(t) = \frac{M_{i-1}(t_i - t)^3}{6h} + \frac{M_i(t - t_{i-1})^3}{6h} + \left(\frac{y_{i-1}}{h} - \frac{M_{i-1}h}{6}\right)(t_i - t) + \left(\frac{y_i}{h} - \frac{M_ih}{6}\right)(t - t_{i-1})$

Interior system:  $\frac{h}{6}M_{i-1} + \frac{2h}{3}M_i + \frac{h}{6}M_{i+1} = d_{i+1} - d_i$ ,  $i = 1, \dots, n - 1$ , plus endpoint equations. Natural:  $M_0 = M_n = 0$ ; coefficient matrix symmetric tridiagonal diagonally dominant PD; Thomas solve  $O(n)$ .

5. CUBIC SPLINES & SPARSE MATRICES

(CONT.)

Sparse structure

Sparsity = #zeros/(mn); density = nnz/(mn). Diagonal density  $\approx 1/n$ ; tridiagonal  $\approx 3/n$ . Hidden sparse product:  $(I + BB^T)x = x + B(B^T x)$ ,  $B \in \mathbb{R}^{n \times r}$ ,  $r \ll n$ . Store/operate on nonzeros. Fill-in = new nonzeros during factorisation; PAQ = LU can reduce fill-in. For banded  $A = LU$  without disruptive pivoting, L/U retain lower/upper bandwidth; SPD  $A = R^T R$  gives R same upper bandwidth.

1-D boundary-value finite difference

$-u'' = f$  on  $(0, 1)$ ,  $u(0) = \alpha$ ,  $u(1) = \beta$ ,  $x_j = j\Delta x$ ,  $\Delta x = \frac{1}{n+1}$ .  $-\frac{U_{j+1} - 2U_j + U_{j-1}}{(\Delta x)^2} = f(x_j)$ ,  $j = 1, \dots, n$ ;  $U_0 = \alpha$ ,  $U_{n+1} = \beta$ ; error  $O((\Delta x)^2)$  under sufficient smoothness. Matrix tridiagonal Toeplitz SPD with diagonal 2 and off-diagonal  $-1$ ; RHS endpoints add  $\alpha, \beta$ .

6. FOURIER ANALYSIS & FFT

Periodic and trigonometric form

$g(t + T) = g(t)$ ,  $f = \frac{1}{T}$ ,  $\cos(\omega t)$ :  $T = \frac{2\pi}{\omega}$ . Harmonic k: frequency  $k/T$ , angular frequency  $2\pi k/T$ .  $g_n(t) = a_0 + \sum_{k=1}^n \left[ a_k \cos\left(\frac{2\pi k t}{T}\right) + b_k \sin\left(\frac{2\pi k t}{T}\right) \right]$ . Over any full period (shown on  $[-T/2, T/2]$ ):  $a_0 = \frac{1}{T} \int_{-T/2}^{T/2} g(t) dt$ ;

$a_k = \frac{2}{T} \int_{-T/2}^{T/2} g(t) \cos\left(\frac{2\pi k t}{T}\right) dt$ ;  $b_k = \frac{2}{T} \int_{-T/2}^{T/2} g(t) \sin\left(\frac{2\pi k t}{T}\right) dt$ . Orthogonality: equal nonzero sine/sine or cosine/cosine integral =  $T/2$ ; unequal or sine/cosine = 0.

***If periodic g is piecewise differentiable: Fourier series converges to g at continuity points and to  $[g(t+) + g(t-)]/2$  at jumps.*** Gibbs oscillations persist near jumps.

Complex Fourier series

$g(t) = \sum_{k \in \mathbb{Z}} c_k e^{i2\pi k t/T}$ ,  $c_k = \frac{a_k - ib_k}{2}$ ,  $c_{-k} = \frac{a_k + ib_k}{2}$ ;  $c_k = \frac{1}{T} \int_{-T/2}^{T/2} g(t) e^{-i2\pi k t/T} dt$ ;

$\text{Co} = \text{ao}$ . ***Pointwise equality stated at continuity points.***

DFT / inverse / FFT

$\omega = e^{i2\pi/n}$ . For  $j, k = 0, \dots, n - 1$ :  $\hat{y}_k = \sum_{j=0}^{n-1} y_j e^{-i2\pi j k/n} = \sum_{j=0}^{n-1} y_j \omega^{-jk}$ ;

$y_j = \frac{1}{n} \sum_{k=0}^{n-1} \hat{y}_k e^{i2\pi j k/n}$ . Direct  $O(n^2)$ ; FFT  $O(n \log_2 n)$ .

$y_j \in \mathbb{R}$ :  $\hat{y}_{n-k} = \overline{\hat{y}_k}$ ,  $\hat{y}_0 \in \mathbb{R}$ ,  $\hat{y}_{n/2} \in \mathbb{R}$  ( $n$  even).

Fourier coefficients from equally spaced data

$t_j = j\Delta t$ ,  $\Delta t = T/n$ ,  $y_j \approx g(t_j)$ ,  $p = \lfloor n/2 \rfloor$ .  $A_0 = \frac{\hat{y}_0}{n}$ ,  $A_k = \frac{2\text{Re}(\hat{y}_k)}{n}$ ,  $B_k = -\frac{2\text{Im}(\hat{y}_k)}{n}$ . If  $n = 2p$  even:  $k = 1, \dots, p - 1$  use ordinary form;  $A_p = \hat{y}_p/n$ ,  $B_p = 0$ . If  $n = 2p + 1$  odd: ordinary form through  $k = p$ . Then  $y_j = A_0 + \sum_k [A_k \cos(2\pi k t_j/T) + B_k \sin(2\pi k t_j/T)]$ .

**De-trending and spectrum**

For  $Y_i \approx x_1 + x_2 t_i + g(t_i)$ ,  $g$  periodic: least-squares fit  $x_1 + x_2 t$  and define  $y_i = Y_i - (x_1 + x_2 t_i)$  before DFT. Power spectrum  $p_k = \frac{|\hat{y}_k|^2}{n}$  plotted against frequency  $k/T$ ; inspect  $k = 0, \dots, \lfloor n/2 \rfloor$  for real data. Large  $p_k$  identify dominant periodic components; period =  $T/k$  ( ***$k > 0$*** ).

7. IMPLIED VOLATILITY & NONLINEAR EQUATIONS

**Black-Scholes call and implied volatility**  
 $\tau = T - t > 0$ .  $c = SN(d_1) - Ke^{-r\tau}N(d_2)$ ;  
 $d_1 = \frac{\ln(S/K) + r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau}$ ,  $d_2 = d_1 - \sigma\sqrt{\tau}$ ;  $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\xi^2/2} d\xi$ .  
Put-call parity:  $c + Ke^{-r\tau} = p + S$ .  
**Require  $S > 0$ ,  $K > 0$ ,  $\sigma > 0$ ,  $\tau > 0$ ; model assumptions in Topic 10.**  
Implied volatility:  $f(\sigma) = c(\sigma) - c_{\text{mkt}} = 0$ ,  $f'(\sigma) = SN'(d_1)\sqrt{\tau} > 0$ .  
Existence and uniqueness when  $\max(S - Ke^{-r\tau}, 0) \leq c_{\text{mkt}} \leq S$ .  
 $f(0^+) = \max(S - Ke^{-r\tau}, 0) - c_{\text{mkt}}$ ,  $f(\infty) = S - c_{\text{mkt}}$ .

**Taylor and finite differences**  
If  $f \in C^{n+1}[a, b]$ ,  $x, x+h \in [a, b]$ :  $f(x+h) = \sum_{k=0}^n \frac{h^k f^{(k)}(x)}{k!} + O(h^{n+1})$ .

$f'(x) = \frac{f(x+h) - f(x)}{h} + O(h) = \frac{f(x) - f(x-h)}{h} + O(h)$  **Require  $f \in C^2$  and shifted point in  $[a, b]$ .**  
 $f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$  **Require  $f \in C^3$  and  $x \pm h \in [a, b]$ .**  
 $f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2)$  **Require  $f \in C^4$  and  $x \pm h \in [a, b]$ .**  
Machine-rounding balance:  
 $O(h) + O(\epsilon h) : h \asymp e^{1/2}$ ,  $O(h^2) + O(\epsilon h) : h \asymp e^{1/3}$ ,  $O(h^2) + O(\epsilon h^2) : h \asymp e^{1/4}$ .  
 **$h$  must be representable so  $x \pm h$  differs from  $x$ .**

**Scalar iterations**  
 $e^{(k)} = x^{(k)} - x^*$ ,  $\beta = \lim_{k \rightarrow \infty} \frac{e^{(k+1)}}{(e^{(k)})^\beta}$ ; for  $v=1$  require  $|\beta| < 1$ .  
Newton:  $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$ . Quadratic locally; **require  $f(x^*)=0$ ,  $f$  and  $f'$  continuous near  $x^*$ ,  $f' \neq 0$  there, initial guess sufficiently close**; asymptotic ratio  $\rightarrow |f''(x^*)|/(2|f'(x^*)|)$ .  
Secant:  $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f(x^{(k-1)}) - f(x^{(k)})}$ ; order  $(1+\sqrt{5})/2$  locally.  
**Require two distinct initial iterates, nonzero divided differences, starts sufficiently close to a simple root.**

**Systems / Gauss-Newton**  
 $r: \mathbb{R}^n \rightarrow \mathbb{R}^m$ ;  $J_{ij} = \frac{\partial r_i}{\partial x_j}$ ,  $r(x+d) = r(x) + J(x)d + O(\|d\|^2)$ . For  $m=n$ ,  
Newton-Raphson:  $J(x^{(k)})d^{(k)} = -r(x^{(k)})$ ,  $x^{(k+1)} = x^{(k)} + d^{(k)}$ ;  **$J$  must be nonsingular at each solve; local quadratic convergence under smoothness and a sufficiently close start.**  
For  $m \geq n$  nonlinear least squares  $\min \|r\|_2^2$ :  $J^T J d = -J^T r$ .  
**Local step is uniquely defined when  $J$  has full column rank.**  
 $J_{ij} \approx \frac{r(x+he_j)_i - r(x)_i}{h}$ ,  $J_{ij} \approx \frac{r(x+he_j) - r(x-h e_j)}{2h}$ ; use  $h \approx \epsilon^{1/2}$  forward,  $h \approx \epsilon^{1/3}$  central.

8. NUMERICAL INTEGRATION

**Probability integrals and quadrature**  
PDF  $p$ :  $p \geq 0$ ,  $\int \mathbb{R} p = 1$ .  $E[f(X)] = \int_{-\infty}^{\infty} f(x)p(x) dx$ ,  $P(x) = \int_{-\infty}^x p(\xi) d\xi$ .  
If  $P$  is differentiable and strictly increasing,  $y = P(x)$  gives  $E[f(X)] = \int_0^1 f(P^{-1}(y)) dy$ .  
 $I(f) = \int_a^b f(x) dx$ ,  $Q(f) = \sum w_i f(\xi_i)$ ,  $E = I - Q$ . Degree of  
precision  $m$ : exact for all polynomials degree  $\leq m$  and not all of degree  $m+1$ .  
**Composite rules**  
Grid  $a = x_0 < \dots < x_N = b$ ,  $h_i = x_i - x_{i-1}$ . Trapezoidal:  
 $Q_N^{\text{trap}} = \sum \frac{h_i}{2} (f_{i-1} + f_i)$ . Uniform  $h = (b-a)/N$ :  
 $h[1/2 f_0 + \sum_{i=1}^{N-1} f_i + 1/2 f_N]$ .  
Midpoint:  $Q_N^{\text{mid}} = \sum_{i=1}^N h_i f(\frac{x_{i-1} + x_i}{2})$ .  
 $|E_N^{\text{trap}}| \leq \frac{(b-a)Mh^2}{12}$ ,  $|E_N^{\text{mid}}| \leq \frac{(b-a)Mh^2}{24}$ . **Require  $f \in C^2[a, b]$  for  $O(h^2)$ .** Both have degree 1.  
Composite Simpson, **uniform grid and even  $N$** :  
 $Q_N^S = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + \dots + 4f_{N-1} + f_N]$ . Degree 3;  $|E_N^S| \leq \frac{(b-a)\max f^{(4)}h^4}{2880}$ .  
**Require  $f \in C^4[a, b]$  for  $O(h^4)$ .**  
 $|E_{2N}| \leq \frac{|E_N|}{2^2}$ ,  $r \approx \log_2 \frac{|E_N|}{|E_{2N}|}$ .

**Gauss-Legendre**  
On  $[-1, 1]$ ,  $N$  nodes/weights chosen exact through  $x^{2N-1}$ ; degree  $2N-1$ .  
 $N=2$ :  $\xi = \pm \frac{1}{\sqrt{3}}$ ,  $w=1$ ;  $N=3$ :  $\xi = 0, \pm \sqrt{\frac{3}{5}}$ ,  $w = \frac{8}{9}, \frac{5}{9}, \frac{5}{9}$ .  
 $\int_a^b f(x) dx \approx \frac{b-a}{2} \sum_{i=1}^N w_i f(\frac{a+b}{2} + \frac{b-a}{2} \xi_i)$ .  
**Non-smooth and multidimensional**  
Formal orders are not guaranteed when required derivatives are missing/unbounded. Split at nonsmooth points or change variables to regularise before applying a high-order rule.  
Tensor product:  $\int_{[0,1]^d} f(x, y) dx dy \approx \sum_{i,j} w_i w_j f(\xi_i, \xi_j)$ . A  $d$ -fold  
( $N+1$ )-point rule uses  $(N+1)^d$  points (**curse of dimensionality**).

9. RANDOM NUMBERS & SIMULATION — DISTRIBUTIONS

True random: physical source. Pseudorandom: deterministic, reproducible, statistically random-like. Quasirandom: deterministic, designed for multidimensional uniformity.  
 $X \sim U[a, b]$ :  $p(x) = \frac{1}{b-a}$ ,  $E[X] = \frac{a+b}{2}$ ,  $\text{Var}(X) = \frac{(b-a)^2}{12}$  ( **$a < b$** ); if  
 $U \sim U[0, 1]$ ,  $Y = a + (b-a)U$ .  
 $U \sim U[0, 1]$ ,  $Y = F^{-1}(U) \implies F_Y = F$ . **Use a suitable generalised inverse if strict invertibility fails.**  
 $Y \sim N(\mu, \sigma^2)$ :  $p(y) = \frac{e^{-(y-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi\sigma}}$ ,  $Y = \mu + \sigma Z$  ( **$\sigma > 0$** ). Independent  
sums:  $\Sigma EX_i = \Sigma EX_i$ ;  $\text{Var} \Sigma X_i = \Sigma \text{Var} X_i$ . CLT: iid sums are approximately normal for large  $n$  and finite variance.

9. RANDOM NUMBERS & SIMULATION — MC/QMC

**Monte Carlo**  
 $E[f] = \int_{\mathbb{R}^d} f(x)p(x) dx$ ,  $x_i \stackrel{\text{iid}}{\sim} p$   
 $E_N[f] = \frac{1}{N} \sum_{i=1}^N f(x_i)$   
Estimator is unbiased; RMS error is independent of  $d$ .  
 $e_N[f] = \frac{\sigma(f)}{\sqrt{N}} = O(N^{-1/2})$   
 $e_N[f] \leq \text{tol} \implies N \geq \left(\frac{\sigma(f)}{\text{tol}}\right)^2$   
**Require independent samples and finite  $\text{Var}[f]$ . Variance reduction**  
Control variate: known  $\mu_g = E[g]$ , estimate  $E[f-g] + \mu_g$ ; useful when  $\text{Var}(f-g) \ll \text{Var}(f)$ .  
Antithetic pairs:  **$N$  even**; on symmetric  $\mathbb{R}^d$  pair  $x$  and  $-x$ ; on  $[0, 1]^d$  pair  $x$  and  $1-x$ . Odd component integrates exactly.

Importance sampling:  $q(x) = g(x)p(x)$ ,  $E[f] \approx \frac{1}{N} \sum_{i=1}^N \frac{f(y_i)}{g(y_i)}$ ,  $y_i \sim q$ .  
**Require  $q \geq 0$ ,  $\int q = 1$ , and  $g(x) \neq 0$  wherever  $f(x) \neq 0$ ; finite variance of  $f/g$  under  $q$ .**

**Quasi-Monte Carlo**  
Deterministic points:  
 $Q_N = \frac{1}{N} \sum_{j=1}^N f(x_j)$ ,  $x_j \in [0, 1]^d$   
Koksma-Hlawka:  
 $|\int_{[0,1]^d} f - Q_N| \leq V(f) D_N^*$

**Requires bounded variation  $V(f)$ .** Star discrepancy:  
 $D_N^* = \sup_{y \in [0,1]^d} \left| \prod_{i=1}^d y_i - \frac{\#\{x_j \in [0, y]\}}{N} \right|$ ,  $D_N^* \leq C \frac{(\log N)^d}{N}$   
Constant  $C$  depends on  $d$ .

10. STOCHASTIC INTEGRATION & SDEs — FOUNDATIONS

**Random walk / Wiener process**  
 $t_n = n\Delta t$ ,  $\Delta t = T/N$ :  
 $\Delta X_n = \mu \Delta t + \sigma(\Delta t)^{1/2} Z_n$ ,  $E[X_N - X_0] = \mu T$ ,  $\text{Var}(X_N - X_0) = \sigma^2 T(\Delta t)^{2\gamma-1}$ ;  
iid  $Z_n \sim N(0, 1)$ . Finite nonzero limit requires  $\gamma = \frac{1}{2}$ .  
 $S_{n+1} = S_n \left(1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_n\right)$ .  
Wiener  $W$ :  $W(0)=0$ ; a.s. continuous paths; increments over non-overlapping intervals are independent: for  $a < b \leq c < d$ ,  $W(b) - W(a)$  is independent of  $W(d) - W(c)$ ;  
 $W(t) - W(s) \sim N(0, t-s)$ ,  $W_{j+1} = W_j + \sqrt{\Delta t} Z_j$ .  
**Itô integral / lemma**  
 $\int_0^T f(W(t), t) dW(t) = \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} \int W(t_n, t_n) [W(t_{n+1}) - W(t_n)]$  (**left endpoints; adapted integrand**).  
 $\sum (\Delta W_n)^2 \xrightarrow{T} \int_0^T W dW = \frac{1}{2} W(T)^2 - \frac{1}{2} T$ .  
 $dX = f(X, t) dt + g(X, t) dW$ ,  $X(0) = X_0$ . For  $Y = H(X, t)$ :  
 $dY = (H_t + \frac{1}{2} g^2 H_{xx}) dt + H_x dX$ . **Require  $H$  continuously differentiable in  $t$  and twice continuously differentiable in  $x$ .**

10. STOCHASTIC INTEGRATION & BLACK-SCHOLES

**Geometric Brownian motion**  
 $dS/S = \mu dt + \sigma dW$ ,  **$S_0 > 0$ ,  $\sigma > 0$** .  $S(t) = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right\}$ ;  
 $\ln\left(\frac{S(t)}{S_0}\right) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right)t, \sigma^2 t\right)$ ;  $S(t)$  is lognormal and positive.  
**Black-Scholes PDE and conditions**  
 $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$ ,  $S > 0$ ,  $0 < t < T$ . Delta hedge uses  $f = V_S$  short shares. **Assumptions: GBM underlying; constant known  $r, \sigma$ ; no dividends, transaction costs, taxes or arbitrage; continuous trading and short selling allowed.**  
 $c(S, T) = \max(S - K, 0)$ ,  $c(0, t) = 0$ ,  $c \sim S - Ke^{-r(T-t)}$ ,  $p(0, t) = Ke^{-r(T-t)}$ ,  $p \rightarrow 0$ .  
Put-call parity:  $S + p = c + Ke^{-r(T-t)}$  (**European options, same  $K, T$** ).

**Log-price/time transformation**  
 $y = \ln(S/K)$ ,  $\tau = T - t$ ,  $V(S, t) = U(y, \tau)$ :  
 $-U_\tau + (r - \frac{1}{2}\sigma^2)U_y + \frac{1}{2}\sigma^2 U_{yy} = rU$ ;  $y \in \mathbb{R}$ ,  $\tau \in [0, T]$ .  
 $U(y, 0) = K \max(e^y - 1, 0)$ ,  $U \rightarrow 0$  ( $y \rightarrow -\infty$ ),  $U \sim Ke^y - Ke^{-r\tau}$  ( $y \rightarrow \infty$ );  
 $U(y, 0) = K \max(1 - e^y, 0)$ ,  $U \sim Ke^{-r\tau}$  ( $y \rightarrow -\infty$ ),  $U \rightarrow 0$  ( $y \rightarrow \infty$ ).

11. FDMs FOR BLACK-SCHOLES

**Grid, notation and differences**  
Truncate  $S \in [0, S_{\max}]$ ,  **$S_{\max} > K$  and sufficiently large**.  
 $\Delta S = \frac{S_{\max}}{N_S + 1}$ ,  $S_j = j\Delta S$ ,  $\Delta t = \frac{T}{N_t}$ ,  $t_n = n\Delta t$ .  $V_j^n \approx V(S_j, t_n)$ .  
 $V_t(S_j, t_n) + rS_j V_S(S_j, t_n) + \frac{1}{2}\sigma^2 S_j^2 V_{SS}(S_j, t_n) = rV_j^n$ .  
 $V_t(S_j, t_n) \approx \frac{V_j^n - V_{j-1}^{n-1}}{\Delta t}$  backward,  $O(\Delta t)$ .  
 $V_S(S_j, t_n) \approx \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S}$ ,  $V_{SS}(S_j, t_n) \approx \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2}$ ; each  $O((\Delta S)^2)$ .  
Boundary/terminal values from Topic 10.

**Backward Euler - EXPLICIT scheme**  
 $\rho = \frac{\Delta t}{2(\Delta S)^2}$ ,  $\alpha_j = (\sigma^2 S_j^2 - rS_j \Delta S)\rho$ ,  $\beta_j = 1 - r\Delta t - 2\sigma^2 S_j^2 \rho$ ,  $\gamma_j = (\sigma^2 S_j^2 + rS_j \Delta S)\rho$ .  
 $V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n$ . Accuracy  $O(\Delta t + (\Delta S)^2)$  when stable.  
Sufficient stability:  $\sigma^2 \geq r$ ,  $\frac{\Delta t}{(\Delta S)^2} \leq \min_{1 \leq j \leq N_S} \frac{1}{r(\Delta S)^2 + \sigma^2 S_j^2}$ ; then  
 $\alpha, \beta, \gamma \geq 0$  and  $|\alpha| + |\beta| + |\gamma| \leq 1$ . Under this mesh relation overall order is  $O((\Delta S)^2)$ .  
**Forward Euler - IMPLICIT scheme**  
 $\alpha_j = \frac{j\Delta t}{2}(r - \sigma^2)$ ,  $\beta_j = 1 + r\Delta t + \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2}$ ,  $\gamma_j = -\frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} - \frac{rS_j \Delta t}{2\Delta S}$ . Solve  
tridiagonal  $\alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n = V_j^{n+1}$  with boundary terms  
in RHS, backwards  $n = N_t - 1, \dots, 0$ . Accuracy  $O(\Delta t + (\Delta S)^2)$ ;  
unconditionally stable.  **$\sigma^2 \geq r$  is sufficient for strict diagonal dominance/nonsingularity, not necessary.**  
Factor once if  $r, \sigma$  time-independent; total  $O(N_t N_S)$ .

**Crank-Nicolson - IMPLICIT scheme; Greeks**  
At  $t_{n+1/2}$ :  $V_j^{n+1/2} = \frac{V_j^n + V_j^{n+1}}{2}$ ,  $V_t \approx \frac{V_j^{n+1} - V_j^n}{\Delta t}$ ; use midpoint value  
in all  $S$  terms. Implicit; accuracy  $O(\Delta t)^2 + (\Delta S)^2$ .  
 $\Delta = V_S$ ,  $\Gamma = V_{SS}$ ,  $\Theta = V_r$ ,  $\rho = V_r$ ,  $\text{Vega} = V_\sigma$ .